ZOTAC GTX 980 AMP! Extreme

There are a lot more AMP!(s) now than usual

Price ₹51,000 (+Tax)

ZOTAC generally had just the one AMP! branded card which was basically a factory overclocked variant with a better cooler with certain PCBs having modifications in order to handle the increased power load. This time around the ZOTAC GTX 980s come in a lot more variants with three SKUs under the AMP! brand. So now they have the plain old AMP!, an AMP! Omega with a moderate overclock and lastly the AMP! Extreme which has a significant overclock. Maxwell scales quite well which is why manufacturers are able to bifurcate the high-end segment even further than before based purely on clock speed. We can expect much more variants once you start factoring in everything else that can be customised.

Coming to the ZOTAC GTX 980 AMP! Extreme, we get to see a factory overclock of 165MHz since the stock GTX 980 is clocked at 1126MHz. Such overclocks aren't alien, even with the GTX 780 Ti we'd seen similar increments with certain cards having approximately 170MHz over the stock GPU. Moreover, if you look at the boost clock then the increment ends up being 102MHz over the normal clock speed whereas the top custom GTX 780 Ti would only have a increment of 60-70MHz. This indicates that Maxwell cards can handle load spikes much better than Kepler cards.

The front of the card protrudes significantly so while the metal bracket occupies two expansion slots, the body will take up another and this will make it difficult for linking two cards in SLI if the slots are side by side. Thankfully, most motherboards which support a dual SLI/Crossfire have the two x16 PCIe slots spaced far apart for such cards. The real problem will arise when one tries 3-way or 4-way SLI. 4-way seems impossible without fitting a custom cooling solution while 3-way might be possible with very few motherboards. A new feature that we saw was Light Id, which is a bunch of LEDs on the card that change colour based on the load that the card is handling at that moment. The card switches between green and red which is a little lacklustre when compared to other manufacturers that use RGB LEDs to include a bigger spectrum. On the rear we see a new additions which is labelled as OC+. This feature requires that you connect a microUSB cable(provided with the card) to the internal USB header to allow you to monitor a lot more parameters that what is normally allowed to on most GPU software. These parameters include voltage levels from the external 8-pin power connectors as well as from the PCIe slot, GPU current draw, memory current draw. It even allows for manipulation of some of these parameters.

If you haven't fiddled around with a lot of graphics cards then this might all seem a bit unsettling, and to make matters worse there isn't much documentation to go by when it comes to all this. Exploring the software that comes with the card will help allay some of your concerns. Just don't click on apply if you aren't sure about what you're doing. Since the software connects to a USB port it is likely that the monitoring and manipulative abilities would be locked to that particular interface. So using any other software will only grant to access to features that are made available by NVIDIA's API. Everything else will require the FireStorm utility.

Performance wise the card does a lot better than everybody else. 3DMark Firestrike score for the ZOTAC GTX 980 AMP! Extreme exceeded that of the stock card by over a thousand points. The FPS count in Bioshock Infinite was a good 7 frames/minute greater that a stock GTX 980. And this difference was maintained in all other gaming benchmarks by a margin of 4-9 FPS. The same goes for OpenCL compute scores which were a good 300 points ahead of stock. So across the board, this card does way better than every card that we have tested so far which is why it gets the Best Performer award in the graphics cards category.

Mithun Mohandas

80

Performance................ 90
Value 75
Build 76

Specifications

Chipset: GM204; **Base clock:** 1291MHz; **Memory clock:** 1800MHz; **CUDA cores:** 2048; **Texture Units:** 128; **ROPs:** 64; **Manufacturing process:** 28nm, PCIe 3.0,4096 x 2160 digital resolution support, 4 GB Memory; **DirectX support:** 12; **OpenGL support:** 4.4; **Power Connectors:** 8Pin + 8Pin; **TDP:** 165+W; **Dimensions (LxWxD):** xxxxxxx mm. **Warranty** - 5 years extended warranty

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